

# Residue Theory

## Complex Analysis

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## Annulus

Laurent Series

Zeroes and  
SingularitiesCauchy's Residue  
Theorem

Jordan's Lemma

Cauchy's Principal  
Value

## Definition

Let  $0 \leq r_1 < r_2$  and let  $z_0 \in \mathbb{C}$ . Then the **annulus around  $z_0$  of inner radius  $r_1$  and outer radius  $r_2$** , denoted by  $A_{r_1}^{r_2}(z_0)$ , is defined to be the set

$$A_{r_1}^{r_2} = \{z \in \mathbb{C} : r_1 < |z - z_0| < r_2\}.$$

The set  $A_0^r(z_0)$  is called the **punctured disc of radius  $r$  around  $z_0$** .

## Definition

Let  $z_0 \in \mathbb{C}$  and let  $\{c_n\}$  be a sequence in  $\mathbb{C}$ . An expression of the form

$$\sum_{n \in \mathbb{Z}} c_n (z - z_0)^n$$

is called a **Laurent series about  $z_0$** . The series is called **convergent** if both the series

$$\sum_{n=0}^{\infty} c_n (z - z_0)^n$$

and

$$\sum_{n=1}^{\infty} c_{-n} \frac{1}{(z - z_0)^n},$$

are convergent. The former and the latter are called the **regular part** (or **analytic part**) and the **principal part** of the Laurent series, respectively.

## Definition

The **annulus of convergence** of a Laurent series  $\sum_{n \in \mathbb{Z}} c_n (z - z_0)^n$  is defined to be the set  $A_r^R$  where

$$r = \limsup \sqrt[n]{|c_{-n}|} \quad \text{and} \quad R = \frac{1}{\limsup \sqrt[n]{|c_n|}}.$$

The justification for the name is given in the following theorem.

### Theorem 1.1

*If  $\{c_n\}$  is a sequence in  $\mathbb{C}$  and  $A$  the annulus of convergence of a Laurent series, then the Laurent series  $\sum_{n \in \mathbb{Z}} c_n (z - z_0)^n$*

- i) converges absolutely at each point of  $A$ .*
- ii) diverges at each point of the set  $\mathbb{C} \setminus A$ .*
- iii) can be both convergent and divergent on  $\partial A$ .*
- iv) converges uniformly on any compact subset  $K \subset A$ .*

## Theorem 1.2

Suppose that  $f$  is analytic in the annulus  $A_{r_1}^{r_2}(z_0)$ . Then the representation

$$f(z) = \sum_{n \in \mathbb{Z}} a_n (z - z_0)^n,$$

called the **Laurent series of  $f$  about  $z_0$** , is valid throughout the annulus. Furthermore, the **Laurent coefficients** are given by

$$a_n = \frac{1}{2\pi i} \int_{\gamma} d\zeta \frac{f(\zeta)}{(\zeta - z_0)^{n+1}},$$

where  $\gamma$  is any simple closed contour contained in the annulus that makes a complete counterclockwise revolution about the point  $z_0$ .

## Theorem 1.3

*The Laurent series of  $f$  about  $z_0$  (in case it is valid) is unique.*

## Theorem 1.4 (Laurent's Theorem)

Suppose  $f \in \mathcal{H}(A_{\rho_1}^{\rho_2}(z_0))$ , then

$$f(z) = \sum_{n \in \mathbb{Z}} c_n (z - z_0)^n \quad \text{for } z \in A_{\rho_1}^{\rho_2}(z_0).$$

Moreover, the coefficients  $c_n$  are given by

$$c_n = \frac{1}{2\pi i} \int_{\Gamma} dz \frac{f(z)}{(z - z_0)^{n+1}},$$

where  $\Gamma$  is the positively oriented circle of radius  $\rho \in (\rho_1, \rho_2)$  and centre  $z_0$ .

## Definition

Let  $\Omega$  be a region, let  $f \in \mathcal{H}(\Omega)$ , and let  $z_0 \in \Omega$ . For  $m \in \mathbb{N}$ , we say that  $z_0$  is a **zero of order  $m$**  of  $f$  if there is another function  $g \in \mathcal{H}(\Omega)$  defined in a neighbourhood  $U$  of  $z_0$  such that  $g(z_0) \neq 0$  and

$$f(z) = (z - z_0)^m g(z) \quad \text{for } z \in U.$$

If  $m = 1$ , we call  $z_0$  a **simple zero** of  $f$ . The zero  $z_0$  is called **isolated** if there is a neighbourhood of  $z_0 \in \Omega$  such that  $z_0$  is the only zero of  $f$  in this neighbourhood.

## Theorem 1.5

Let  $\Omega$  be a region containing a point  $z_0$  and suppose that  $f$  is in  $\mathcal{H}(\Omega)$  with  $f(z_0) = 0$ . Then exactly one of the following two assertions holds:

- i)  $f$  is identically zero in a neighbourhood of  $z_0$ .
- ii)  $z_0$  is an isolated zero of  $f$ .

Moreover, if (ii) holds, then there are a natural  $m$ , a real number  $r > 0$ , and a holomorphic function  $\lambda$  on  $B_r(z_0)$  such that  $\lambda(B_r(z_0)) \cap \{0\} = \emptyset$  and

$$f(z) = (z - z_0)\lambda(z) \quad \text{for } z \in B_r(z_0).$$

Consequently,  $z_0$  is a zero of order  $m$  and we have

$$f^{(n)}(z_0) = 0 \quad n = 1, \dots, m-1, \quad f^{(m)}(z_0) \neq 0.$$

## Theorem 1.6

*A nonzero holomorphic function defined on a region has isolated zeroes.*

### Theorem 1.7 (Identity Principle)

*Suppose that  $f, g$  are holomorphic functions on a region  $\Omega$ . If  $(z_n)$  is a sequence of distinct points in  $\Omega$  with  $f(z_n) = g(z_n)$  for all  $n$  and  $z_n \rightarrow z_0 \in \Omega$ , then  $f = g$  on  $\Omega$ .*

## Definition

A point  $z_0 \in \mathbb{C}$  is called a **singularity** of a function  $f$  if  $f$  fails to be differentiable at  $z_0$  but is differentiable at some point in every neighbourhood of  $z_0$ . A singularity of  $f$  is said to be an **isolated singularity** if, in addition, there is a punctured disc  $A_0^\varepsilon(z_0)$  around  $z_0$  throughout which  $f$  is differentiable. An isolated singularity  $z_0$  of  $f$  is called a

- **removable singularity** if there exists a finite  $\lim_{z \rightarrow z_0} f$ .
- **pole** if  $\lim_{z \rightarrow z_0} f = \infty$ .
- **essential singularity** if the limit  $\lim_{z \rightarrow z_0} f$  does not exist.

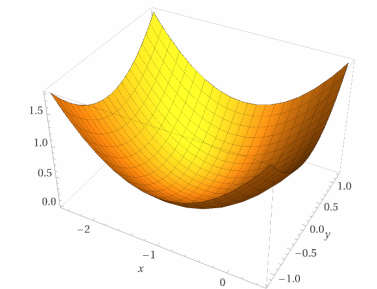


Figure 1: Near a removable singularity,  $|f|$  is bounded

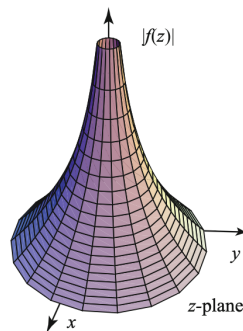


Figure 2: Near a pole,  $|f|$  tends to infinity

## Example 1.8

*For the following functions, determine their singularities and classify them.*

a)  $f(z) = \frac{\sin z}{z}$

b)  $f(z) = \frac{z}{1+z}$

c)  $f(z) = \exp \frac{1}{z}$

**Theorem 1.9**

*Let  $h$  be holomorphic on  $B_R(z_0)$  with a zero of order  $m \geq 1$  at  $z_0$ . If  $p \leq m$  is an integer, then  $f(z) = h(z)/(z - z_0)^p$  has a removable singularity at  $z_0$ .*

### Theorem 1.10 (Riemann's Principle of Removable Singularities)

*Suppose  $f$  is holomorphic on  $0 < |z - z_0| < R$ . Then  $f$  has a removable singularity at  $z_0$  if and only if*

$$\lim_{z \rightarrow z_0} (z - z_0) f(z) = 0.$$

### Theorem 1.11 (Characterisation of Removable Singularities)

Suppose that  $f$  is holomorphic on  $0 < |z - z_0| < R$ . Then the following are equivalent:

- i)  $f$  can be redefined at  $z_0$  to be holomorphic there.
- ii)  $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$  for  $0 < |z - z_0| < R$ .
- iii)  $f$  has a removable singularity at  $z_0$ .
- iv)  $\lim_{z \rightarrow z_0} |f(z)|$  exists and is finite.
- v)  $f$  is bounded on a neighbourhood of  $z_0$ .
- vi)  $\lim_{z \rightarrow z_0} f(z) = 0$ .

## Definition

Let  $z_0$  be a pole of a holomorphic function  $f$  on  $A_0^R(z_0)$ . The **order** of the pole of  $f$  at  $z_0$  is defined as the order of the zero of  $1/f$  at  $z_0$ .

## Theorem 1.12 (Characterisation of Poles)

Let  $m \in \mathbb{N}$ . Let  $R > 0$  and suppose that  $f$  is holomorphic on the punctured disc  $A_0^R(z_0)$ . Then the following conditions are equivalent:

- i)  $f$  has a pole of order  $m$  at  $z_0$ .
- ii) There is an  $r > 0$  and there is a nowhere vanishing holomorphic function  $\Psi \in \mathcal{H}(B_r(z_0))$  such that

$$f(z) = \frac{\Psi(z)}{(z - z_0)^m}$$

on  $A_0^R(z_0)$

- iii) There exists a complex number  $\alpha \neq 0$  such that

$$\lim_{z \rightarrow z_0} (z - z_0)^m f(z) = \alpha.$$

- iv) There exist  $r > 0$  and complex numbers  $a_n$  for  $n \geq -m$  such that  $a_{-m} \neq 0$  and

$$f(z) = \sum_{n=-m}^{\infty} a_n (z - z_0)^n$$

for all  $z \in A_0^r(z_0)$ .

## Theorem 1.13 (Characterisation of Singularities via Laurent Series)

Suppose  $f \in \mathcal{H}(\Omega \setminus \{z_0\})$ . Let

$$f(z) = \sum_{n \in \mathbb{Z}} a_n (z - z_0)^n$$

be the Laurent series expansion of  $f$  about  $z_0$ , which is valid in some annulus  $A_0^R(z_0)$ . Then

- i)  $z_0$  is a removable singularity if and only if it has no principal part in its Laurent series, i.e.,  $a_n = 0$  for all  $n < 0$ .
- ii)  $z_0$  is a pole of order  $m \in \mathbb{N}$  if and only if the principal part of its Laurent series has finitely many terms, i.e.,  $a_{-m} \neq 0$  for some  $m > 0$ .
- iii)  $z_0$  is an essential singularity if and only if the principal part of its Laurent series has infinitely many terms, i.e.,  $a_n \neq 0$  for infinitely many  $n < 0$ .

### Theorem 1.14 (Casorati-Weierstrass)

*Suppose that  $f$  is holomorphic on a punctured disc  $B'_R(z_0)$ . Then  $z_0$  is an essential singularity of  $f$  if and only if the following two conditions hold:*

- i) *There is a sequence  $(z_n)$  in  $B'_R(z_0)$  such that  $z_n \rightarrow z_0$  and  $|f(z_n)| \rightarrow \infty$  as  $n \rightarrow \infty$ .*
- ii) *For any complex number  $\alpha$ , there is a sequence  $(z_n)$  in  $B'_R(z_0)$  (that depends on  $\alpha$ ) such that  $z_n \rightarrow z_0$  and  $f(z_n) \rightarrow \alpha$  as  $n \rightarrow \infty$ .*

## Definition

Suppose that  $f \in \mathcal{H}(B_R(z_0))$  for some  $R > 0$ . Then

- i) a **removable singularity at  $\infty$**  if  $f(1/z)$  has a removable singularity at 0;
- ii) a **pole of order  $m$  at  $\infty$**  if  $f(1/z)$  has a pole of order  $m$  at 0;
- iii) an **essential singularity at  $\infty$**  if  $f(1/z)$  has an essential singularity at 0.

When  $f$  has a removable singularity at  $\infty$ ,  $\lim_{z \rightarrow \infty} f(z)$  exists. When  $\lim_{z \rightarrow \infty} f(z) = 0$ , we say that  $f$  has a **zero at  $\infty$** .

## Definition

Let  $f \in \mathcal{A}(A_0^R(z_0))$  with  $z_0 \in \mathbb{C}$ . The **residue of  $f$  at  $z_0$** , denoted by  $\text{Res}(f; z_0)$  (or simply,  $\text{Res}_{z_0} f$ ) is the  $-1$ th Laurent coefficient

$$\text{Res}(f; z_0) = \frac{1}{2\pi i} \oint_{B_\rho(z_0)} dz f(z) = a_{-1},$$

where  $\rho \in (0, R)$ .

### Theorem 1.15 (Cauchy's Residue Theorem)

Let  $\Gamma$  be a simple closed contour. Suppose that  $f \in \mathcal{A}(\Gamma \cup \Gamma^\circ \setminus \{z_k\}_{k=1}^n)$  with  $\{z_k\}_{k=1}^n \subset \Gamma^\circ$ . Then

$$\oint_{\Gamma} f = 2\pi i \sum_{k=1}^n \operatorname{Res}(f; z_k).$$

**Theorem 1.16**

*If a function  $f$  can be written as*

$$f(z) = \frac{p(z)}{q(z)},$$

*where  $p$  and  $q$  are analytic at  $z_0$ ,  $p(z_0) \neq 0$ , and  $q$  has a simple zero at  $z_0$ , then*

$$\operatorname{Res}(f; z_0) = \frac{p(z_0)}{q'(z_0)}.$$

**Theorem 1.17**

*Suppose that  $z_0$  is a pole of order  $m \geq 1$  of  $f$ . Then we have*

$$\operatorname{Res}(f; z_0) = \lim_{z \rightarrow z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)].$$

## Theorem 1.18

*Suppose that  $0$  is an isolated singularity of an even function  $f$  that is analytic on  $\mathbb{C} \setminus \{0\}$ . Then  $\text{Res}(f, 0) = 0$ .*

**Theorem 1.19**

*If  $f$  has a simple pole at  $z_0$  and  $g$  is analytic at  $z_0$ , then*

$$\operatorname{Res}(fg; z_0) = g(z_0)\operatorname{Res}(f; z_0).$$

### Theorem 1.20 (Jordan's Lemma)

Let  $C_R$  be a semicircle of radius  $R$  in the upper half plane and centred at the origin. Let  $f$  be a function that tends uniformly to zero faster than  $1/|z|$  for  $\text{Arg } z \in [0, \pi]$  as  $|z| \rightarrow \infty$ . Let  $\alpha$  be a nonnegative real number. Then

$$\lim_{R \rightarrow \infty} I_R := \lim_{R \rightarrow \infty} \int_{C_R} dz e^{i\alpha z} f(z) = 0.$$

## Remark

*So far we have discussed only integrals of functions that have no singularities on the contour. Let us now investigate the consequences of the presence of singular points on the contour. Consider the integral*

$$\int_{\mathbb{R}} dx \frac{f(x)}{x - x_0},$$

*where  $x_0$  is a real number and  $f$  is analytic at  $x_0$ . To avoid  $x_0$  (which causes the integrand to diverge) we bypass it by indenting the contour as shown below and denoting the new contour by  $C_u$ . The contour  $C_0$  is simply a semicircle of radius  $\varepsilon$ . For the contour  $C_u$ , we have*

$$\int_{C_u} dz \frac{f(z)}{z - x_0} = \int_{-\infty}^{x_0 - \varepsilon} dx \frac{f(x)}{x - x_0} + \int_{x_0 + \varepsilon}^{\infty} dx \frac{f(x)}{x - x_0} + \int_{C_0} dz \frac{f(z)}{z - x_0}.$$

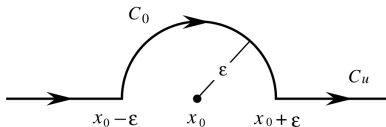


Figure 3: The contour  $C_u$  avoids  $x_0$

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## Definition

The **Cauchy's principal value** of the integral

$$\int_{\mathbb{R}} dx \frac{f(x)}{x - x_0},$$

denoted by P.V. is defined as

$$\text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} = \lim_{\varepsilon \rightarrow 0} \left( \int_{-\infty}^{x_0 - \varepsilon} dx \frac{f(x)}{x - x_0} + \int_{x_0 + \varepsilon}^{\infty} dx \frac{f(x)}{x - x_0} \right)$$

## Remark

*The integral over the semicircle is calculated directly, yielding*

$$\int_{C_0} dz \frac{f(z)}{z - x_0} = -i\pi f(x_0).$$

*Therefore*

$$\int_{C_u} dz \frac{f(z)}{z - x_0} = \text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} - i\pi f(x_0).$$

*On the other hand, if  $C_0$  is taken below the singularity on a contour  $C_d$ , say, we obtain*

$$\int_{C_d} dz \frac{f(z)}{z - x_0} = \text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} + i\pi f(x_0).$$

## Remark

We see that the contour integral depends on how the singular point  $x_0$  is avoided. However, the principal value, if it exists, is unique. To calculate this principal value we close the contour by adding a large semicircle to it as before, assuming that the contribution from this semicircle goes to zero by Jordan's lemma. The contour  $C_u$  [respectively,  $C_d$ ] is replaced by a closed contour, and the value of the integral will be given by the residue theorem. We therefore have

$$\text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} = i\pi f(x_0) + 2\pi i \sum_j \text{Res} \left( \frac{f(z)}{z - x_0}; z_j \right),$$

$$\left[ \text{respectively } \text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} = -i\pi f(x_0) + 2\pi i \sum_j \text{Res} \left( \frac{f(z)}{z - x_0}; z_j \right) \right],$$

where  $\{z_j\}_{j=1}^n$  are the poles of  $f(z)$ .